

Bowen Groff

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Engineering

Orlando, FL

Aug 2024 – May 2028

EXPERIENCE

GPU Software Engineer Intern

January 2026 – May 2026

AMD

Orlando, FL

- Ported **10+** existing **HIP**-based GPU kernels into a testbench suite within the **ROCm** stack for the **AMD Instinct** series.
- Implemented and benchmarked **async load/store**, **atomic**, **launch latency**, and **throughput-unrolling** kernel tests, using **C++20 compile-time optimization** techniques to improve kernel performance by **up to 73%** in targeted cases.
- Implemented **concurrent kernel execution** across streams, increasing performance by **n-fold**, depending on stream count.
- Maintained **HPC server infrastructure**, including hardware replacement and validation of **NVIDIA H100** GPUs.

Design Director

February 2026 – Present

Knight Hacks

Orlando, FL

- Leading a team of **18 designers** across **UI/UX**, **illustration**, and **graphic design**, delegating work and providing direction.
- Developed the visual identity for **KHIX**, including the brand system, logo, hero artwork, and **15+ merchandise designs**, driving a **3x increase in first-week applicants** compared to the previous year.
- Delivered **dozens of illustrations and graphics** across social media, general body meetings, promotional campaigns, and event materials, reaching a community of **5,000+ Discord members** and **3,000+ Instagram followers**.

Software Engineer Intern

May 2025 – November 2025

L3Harris Technologies

Palm Bay, FL

- Developed a **platform Linux kernel driver** in **C**, enabling bidirectional Ethernet and Fibre Channel message transport between an **FPGA** and userspace over **UDP**, handling **over 810** concurrent message buffers sorted through **8** tables.
- Mapped reserved memory blocks in **custom device tree** with **C++** application, enabling low-latency message handling.
- Received **interrupts** from an FPGA and wrote **custom ioctl macros** to post interrupt details directly to userspace.

Software Engineer Intern

May 2024 – April 2025

L3Harris Technologies

Colorado Springs, CO

- Replaced a legacy CLI-driven workflow with a **React** and **Spring Boot** web application exposing internal tools for deep-space telescope operators, reducing data conversion time by **~90%** and streamlining operator onboarding for **.FITS** file conversion.
- Implemented a **Java** backend that parsed **.ini configurations**, indexed **.FITS** data from network-mounted drives, and automatically filtered files by **telescope**, **date**, and **time**, enabling external program execution via **Spring Boot APIs**.

Software Engineer Intern

May 2023 – August 2023

L3Harris Technologies

Palm Bay, FL

- Designed and trained **three predictive neural network models** (including **multi-layer perceptrons**) using **PyTorch** and **TensorFlow** to analyze **word-frequency features** from a dataset of **<1,000 samples**, achieving **~68%** prediction accuracy.
- Awarded the **L3Harris RISE Award** after presenting to **key stakeholders** for enabling **F-35 ICP failure mode diagnosis** **~99.5% faster**, resulting in an estimated **\$4M** in operational savings and **thousands** of engineering hours saved overall.

PROJECTS

Morphogen | CUDA, C++20, CMake

July 2026 - Present

- Developing a **CUDA/C++** simulation engine for GPU-accelerated mathematical pattern generation, currently supporting **Mandelbrot fractal** generation.
- Implemented a **CUDA kernel** mapping individual pixels to GPU threads to parallelize fractal computation.

Forge | Node.js, PostgreSQL, Discord API, tRPC, Zod, Cron

September 2025 - Present

- Designed and deployed a **Node.js** judging backend structure using **tRPC** and **Zod** for end-to-end API validation, dynamically assigning judges to hacking rooms during a live **~1,000-participant** event with **zero runtime failures**.
- Implemented an automated **cron-based role synchronization pipeline** integrating PostgreSQL and **Discord API**, achieving **~99%** reduction in manual overhead for alumni role management and ensuring consistent permissions enforcement.

SightSpeech | Python, EasyOCR, Gemini API, Flask, MediaPipe

September 2025

- Achieved **1st place out of 113 projects and 452 competitors** for MLH's Best Use of Gemini API at Shell Hacks 2025.
- Built a real-time assistive OCR pipeline using **EasyOCR** for bounding-box word detection and **Gemini API** for post-OCR contextual correction and grouping, achieving **~97% word accuracy** across handwritten text and mathematical expressions.
- Coordinated a split-camera architecture with **MediaPipe** gesture recognition in the JS frontend and **EasyOCR** in the **Python/Flask** backend, bridged via **REST API** to map **4 gestures** to real-time text navigation and **text-to-speech** output.

TECHNICAL SKILLS

Languages: C, C++ (C++20), Java, Python, JavaScript, TypeScript, RTL, ARMv7 Assembly

Systems & Tools: Atlassian Suite, GitHub, Linux, Git, Docker, Jenkins, ROCm, HIP, Figma, Adobe Suite

Frameworks: React, React Native, Next.js, Node.js, Flask, Spring Boot, PyTorch, TensorFlow, Pandas, Keras